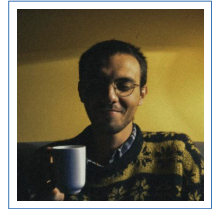


Lorenzo Maniscalco

Curriculum Vitae

Dipartimento di Matematica
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Research interests

Geometric analysis and elliptic PDEs, with focus on spacelike hypersurfaces in Lorentzian manifolds, prescribed mean curvature problems and capillary boundary conditions.

Education

- 2023–present **PhD student, Mathematics**, *Università degli Studi di Torino*, Supervisors: Luciano Mari (Università di Milano), Alessandro Iacopetti, *Expected defense: November 2026*.
- 2021 – 2023 **Master in Mathematics**, *Università degli Studi di Torino*, 110/110 *cum laude* with distinction, Thesis: *Existence and regularity for spacelike hypersurfaces with prescribed mean curvature*.
- 2019 – 2023 **Bachelor in Mathematics**, *Università degli Studi di Torino*, 110/110 *cum laude*, Thesis: *Il teorema della mappa di Riemann*.

Publications

Preprints

- 2026 Gabriele Cora, Alessandro Iacopetti, and Lorenzo Maniscalco. Entire spacelike radial graphs with prescribed mean curvature in the Lorentz–Minkowski space. arXiv:2605.06409, 2026.
- 2025 Lorenzo Maniscalco and Luciano Mari. Prescribing the mean curvature of an achronal hypersurface as a measure: the case of 3D spacetimes. arXiv:2512.17670, 2025.

Talks

- 19/02/2025 **Lorentzian geometry and non-linear electrodynamics**, *Seminario del Dottorato*, Università di Torino.
- 26/02/2025 **Spacelike hypersurfaces with singular mean curvature**, *Seminario MAP, Mathematical Analysis and Probability*, Scuola Normale Superiore, Pisa.
- 16/04/2025 **Prescribing mean curvature in Lorentzian geometry**, *PhD days*, Università di Milano.
- 01/12/2025 **On the mean curvature of spacelike hypersurfaces asymptotic to a light cone**, *Séminaire de clôture du projet “PDEs in interactions”*, Université Libre de Bruxelles, Spa, Belgium.
- 20/01/2026 **Born–Infeld electrodynamics and spacelike hypersurfaces with singular mean curvature**, *PDEs at Grand Paradis*, Cogne, Italy.
- 27/04/2026 **Where electrodynamics meets Lorentzian geometry: the Born–Infeld equation**, *Insalate di Matematica*, Università Milano Bicocca, Milan, Italy.

Teaching Assistantship

- Fall, 2022 **Tutor, Geometria 2 (Bachelor in Mathematics)**, 50 hours, Università di Torino.
- Spring, 2023 **Tutor, Algebra 1 (Bachelor in Mathematics)**, 50 hours, Università di Torino.
- Fall, 2023 **Tutor, Meccanica Razionale (Bachelor in Mathematics)**, 50 hours, Università di Torino.
- Fall, 2023 **Tutor, Geometria e Algebra Lineare 1 (Bachelor in Physics)**, 75 hours, Università di Torino.

Fall, 2024 **Tutor, Algebra 1 (Bachelor in Mathematics)**, 50 hours, Università di Torino.
Spring, 2026 **Tutor, Geometria 1 (Bachelor in Mathematics)**, 32 hours, Università di Torino.

Languages

Italian **Native**
English **C1**
French **B2**

References

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Prof. Alessandro Iacopetti

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